Assignment: Introduction, Material & Methods

Introduction

This section should provide enough background for the reader to understand the purpose and significance of your project. Begin broadly by placing your work in the context of existing research, describe relevant studies, summarize key findings, and explain the rationale behind your project.

Conclude the introduction with a clear thesis statement or hypothesis. This should present a broad, "big picture" idea that proposes a model or system, rather than a specific prediction about an individual experiment. After stating your hypothesis, briefly summarize the experiments you conducted to test it.

As you write, remember that your **Discussion** section will likely expand on the background you present here, connecting your results to prior research. Keep this in mind when drafting your Introduction, include only the background information necessary to set up your project, and reserve more detailed analysis or interpretation for the Discussion.

Prompts for the Introduction

1. What is the overall purpose of this cloning project? Can this purpose be phrased as a thesis statement or a hypothesis? If your hypothesis is correct, then what results would you predict for your experiment?

2. Why is this project important?

3. What background information do you need to know in order to understand the goals of this cloning project? What key concepts are involved?

4. How have the concepts behind this cloning project been applied previously in research? What details about these prior studies should be introduced in this section, before being compared with your research findings and further discussed in your Discussion section?

5. What review articles (from peer-reviewed journals) could you recommend to your readers so that they can learn more about this background?

6. What references need to be cited within the text of your Introduction?

Materials and Methods

Your readers need to know the procedures that were carried out and the materials that were used in order to be able to reproduce the findings of the study. Having reproducible results is key to establishing new findings and laying solid foundations for further research. Thus, each primary research article contains a Materials and Methods section.

Typically, the Materials and Methods section describes a procedure in enough detail for a researcher to buy similar materials and to work out the conditions required to do the experiment. However, you will

notice that it does not describe step-by-step procedures. Rather, it refers the reader to the previously published protocols, describes the modifications made to those published protocols, and provides summaries of what was done. If materials were provided by other researchers, the citations that describe those materials are given.

(Normally, one lists all materials and equipment used, including brand or model, company, each cited where appropriate. However, since you have not been told the source of all your materials, these may be described as "All materials and equipment were provided by the BIOL 341 laboratory." This is the one phrase you may copy exactly in your report.)

The following questions are intended to help you with your draft Introduction and your draft Materials and Methods, but they do not need to be answered in your lab notebook. Read Presenting your work as a journal article and look at the relevant sections of the Generic marking checklist for draft sections and final reports (both in Lab Protocols) for more tips on writing your drafts.

Prompts for Materials and Methods

1. Did you use material that was prepared by other students? (e.g. digested plasmid; genomic DNA fragments) Cite these students. (They should also be thanked in your Acknowledgements.)

2. What procedures have you carried out for your cloning project? The procedures should be subtitled and summarized briefly.

3. What is a generic description of your procedure, and what details can be left out without affecting the results? For example, the volume of your restriction digests would not be expected to influence the results of your restriction analysis. Where would you refer the reader for details about the protocols?

4. What were specific aspects to your procedures that need to be mentioned so that your readers can reproduce the experiments? e.g. restriction digest protocols are standard and can be looked up, but the exact restriction endonucleases used would affect your results and analysis.

5. What in-text citations need to be included?

This assignment is to be done individually. Refer to the grading rubric to ensure that the proper format, style, and content have been included in your draft:

- The page limit for the Introduction is 3-4 pages (double-spaced).
- The page limit for Materials and Methods is about 2 pages (double-spaced) for the experiments to date. (The final page limit will be 2-3 pages.)
- Text should be double-spaced with 1 inch (2.5 cm) margins and in 11 or 12-point font, a professional font like Arial or Times New Roman).
- Your draft should include in-text citations plus end references. These should be in APA format.
- End references may be double-spaced according to proper APA format. To conserve paper, references may instead be single-spaced in 9-pt. font with a single line to separate references.

- This draft is due lab 8 for peer review, and Thursday Nov 12th by midnight or marking by your TAs.
- Submit a PDF file to the course site for marking. This will eventually become part of your final report.